BindFormer Evidence-grouped compression for 3D MRI report generation

(a) End-to-end framework

One MRI sequence · fixed visual-token budget



3D MRI

NeuroVFM
Frozen base + LoRA

$X: N \times 768$

BindFormer projector
Group → read roles → compress

$Z: M \times 4096$

Llama 3.1 · 8B
Frozen

Findings

(b) BindFormer projector

Separate evidence groups; bind complementary attributes through shared support

Full visual memory H

X: all patch features
No pre-pooling

Input adapter
LN + projection
3D PE + sequence embedding

Spatial + sequence context

3D coordinates p_i
Sequence identity m

H

Competitive slot routing
3 iterations · $K+1$ slots

s_1 s_2 s_K bg

A

Soft patch-to-slot assignment
S: learned evidence groups

A

Support geometry (opt.)
$A, p \rightarrow \mu_k, \Sigma_k, \nu_k$

S

Fixed concept memory
UMLS names + short definitions
SapBERT · cached embeddings

Role-wise concept retrieval
Role / sequence candidate mask

e_{kr}

Instance × role queries

F L S M

$q_{kr} = W_q s_k + v_r + U_r e_{kr} + g_{kr}$

F finding L location / laterality
S signal M morphology / extent

Geometry → L / M queries

Q

Global cross-attention
G learned queries attend to H

Global structure + normal context

H: visual keys / values

Support-biased visual read
$\text{score}(q_{kr}, h_i) + \beta_r \log(A_{ik} + \varepsilon)$
Softmax over image patches

F L S M

$b_{kr} = \sum_i \alpha_{kri} W_V h_i$

Role-specific weights
Shared visual source

Shared support $A_{:k}$ across all roles of slot k

Within-slot interaction (opt.)
Role self-attention
Gated read of global tokens

Keep slot-wise role bundles

Final LN + output MLP

Z

$M = G + KR$ · $R = 4$

(c) Training with paired MRI and reports

Auxiliary heads read final Z; removed for inference

Visual information retention

Z K/V **Reconstructor R_ψ**
CA + FFN → 768-d
$\widehat{x}_i$

Queries: p_i + sequence m

LN-MSE

sg(x_i^T) · frozen teacher → $\mathcal{L}_{\text{feat}}$

Clinical phrase alignment

Bundle Z_k → **Pool + project** → u_k → **MIL matching** $\max_k \cos(u_k, v_\ell)$ → $\mathcal{L}_{\text{text}}$

Report snippets → **Text encoder** Frozen · cached v_ℓ → v_ℓ

Report generation

Z → **Frozen Llama** Teacher forcing → $\mathcal{L}_{\text{RG}}$ Token CE

$y_{<t}$

Reference Findings y → $\mathcal{L}_{\text{RG}}$

$$\mathcal{L} = \mathcal{L}_{\text{RG}} + \lambda_f \mathcal{L}_{\text{feat}} + \lambda_s \mathcal{L}_{\text{text}}$$

Gradients reach the projector; Llama parameters stay frozen

Training schedule

1 Projector warm-up
Projector LR ↑ · LoRA frozen

→ **2 LoRA warm-up**
LoRA LR ↑ · projector LR held constant

→ **3 Joint training**
Update projector + visual LoRA

→ Forward / inference ⇢ Training only ▭ Opt.: optional module

Slots are evidence groups; the background slot emits no token.